A Deep Learning Framework With Multimodal Molecular Representation For Molecular Property Prediction

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***Abstract*—Molecular property prediction is a basic research problem in the field of drug design and discovery. Effective molecular representation can help models accurately predict the properties of molecules. To capture the physical-chemical properties of molecular fingerprint, the functional group information of molecular fragment and the structural features of molecular graph. This paper first combined molecular fragment and molecular graph representation to extract local and global features of molecules. Then, molecular fingerprint was added to the graph representation to achieve a combination of multi-modal representation. And applied the method to establish a molecular property prediction model, called PFG. To evaluate the PFG model, experiments were conducted on 10 benchmark datasets applying the receiver operating characteristic curve (ROC-AUC), root mean square error (RMSE) and mean absolute error (MAE) evaluation indicators. Compared with models based only on fingerprint or graph representation, the model achieved better prediction results on 9 datasets. The effectiveness of the combined molecular representation method proposed in this paper was verified. Compared with SVM, GBDT, RF, XGBoost, GAT, Attention FP, DMPNN, FP-GNN and FraGAT models, the effectiveness of the proposed method is proved.**

Keywords—molecular property prediction, deep learning, molecular representation, graph convolutional networks

I. INTRODUCTION

Molecular property prediction is one of the important tasks in the field of computer-aided drug design. With the increasing use of deep learning (DL) methods to explore molecules, DL has many applications and achieved a series of results in predicting molecular properties. Effective molecular representation can help models accurately predict the properties of molecules. There are generally 3 methods for representation of molecules. The first is a method based on SMILES, including SA-BiLSTM [1], Molecule-Transformer [2], and SMILES-BERT [3], which all use SMILES strings as input and use natural language models as the basic model. The second method is based on molecular fingerprint, including ECFP [4] and Mol2vec [5], which simplify and abstract molecules by extracting structural features of molecules to generate bit vectors. The third method is based on molecular graph, including MPNN [6] and DMPNN [7]. In the molecular graph representation, atoms are represented as nodes of the graph, and the physical and chemical properties of the atoms are used as node features. Chemical bonds are represented as edges of the graph, and the properties of the bonds are used as edge features [8]. However, SMILES is a one-dimensional linearization of molecular structures. It is difficult to learn the original structural information of molecules based on strings and to easily add other features, making its application in DL methods relatively inconvenient. Molecular fingerprint is difficult to reflect the importance of different substructures and difficult to train in an end-to-end manner [9]. For molecular graph representation, a large amount of training data is usually required due to the complex architecture of graph neural networks. This limits the generalization ability of GNN when data is sparse, and it may be difficult to learn a robust graph representation from a limited dataset [10].

The molecular information that can be captured by a single molecular representation is limited, and different molecular representations have different focuses. Therefore, many researchers had begun to try to combine multiple molecular

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representations to obtain new molecular representation forms. Rahaman et al. [11] proposed a general DL based framework to combine different types of molecular fingerprint to improve prediction accuracy, showing that the combination of molecular fingerprint has better prediction results than a single molecular fingerprint. Rifaioglu et al. [12] found that based on image or fingerprint classifiers showed opposite trends in predicting the properties of protein families. This finding suggests that the information captured by different forms of molecular representations may be different and potentially complementary. Zhang et al. [13] proposed the FraGAT model by combining molecular fragment with molecular graph representation, which showed that there is a correlation between molecular fragment and molecular properties. Cai et al. [14] proposed FP-GNN by integrating molecular graph and molecular fingerprint, which showed excellent prediction performance on multiple public benchmark datasets and was better than a single molecular representation. Therefore, representation methods that integrate multiple molecular representations are feasible and can improve the predictive performance of the model.

On this basis, this paper constructed embedding vectors based on molecular fragment and atomic properties respectively. The local and global features of molecules were extracted by GCN by combining molecular fragment and molecular graph representation. And further proposed a molecular representation method that combines mixed molecular fingerprint and graph representation. Then a DL model (PFG) was built based on this method for molecular property prediction. The PFG model was evaluated on 10 benchmark datasets, comparing PFG with models based on molecular fingerprint or molecular graph representation methods. The PFG model achieved better prediction results on the tasks of 9 datasets. It was verified that combining molecular fingerprint and graph can improve the prediction performance of the model. Then, PFG was compared with SVM, GBDT, RF, XGBoost, GAT, Attention FP, DMPNN, FraGAT, and FP-GNN models. The PFG model achieved better prediction results on the tasks of 9 datasets and suboptimal prediction results on the task of 1 dataset. It was proved that the PFG model has better prediction performance. This paper combined molecular fingerprint, molecular graph, and molecular fragment to improve the performance of the model, which provided an effective molecular representation method for molecular property prediction. The paper is structured as follows, with the first section providing a background on molecular representation methods. The second part is an introduction to the dataset and model framework used. The third part is the comparison experiment of the model. The fourth part is the summary of the paper.

II. Materials and Methods

A. Datasets

This paper uses 10 benchmark datasets for molecular property prediction to evaluate the performance of the prediction model. The datasets used are BBBP, SIDER, ClinTox, BACE, and Tox21 datasets for classification tasks and Freesolv, ESOL, Lipo, QM7, and QM8 datasets for regression tasks. The information for each baseline dataset is shown in Tab. 1. All benchmark datasets are divided into training set, validation set, and test set into 8:1:1 by random splitting. For multi-task datasets, each sub-task is trained separately in single-task mode, and the average of all sub-tasks is taken as the multi-task dataset result. The evaluation indicators used in this paper are as follows:

TABLE I. Baseline dataset information

Type	*Category*	*Dataset*	*Tasks*	*Compounds*	*Metric*
Classifi-cation	Physiology and toxicity	BBBP	1	2039	ROC-AUC
		SIDER	27	1427	ROC-AUC
		ClinTox	2	1478	ROC-AUC
		Tox21	12	7831	ROC-AUC
	Biological activity	BACE	1	1513	ROC-AUC
Regres-sion	Physical chemistry	Freesolv	1	642	RMSE
		ESOL	1	1128	RMSE
		Lipo	1	4200	RMSE
	Quantum mechanics	QM7	1	6830	MAE
		QM8	12	21786	MAE

1)ROC-AUC: ROC-AUC is a commonly used evaluation index used to evaluate the performance of binary classification models. It ranges from 0.5 to 1.0, the higher the value, the better the performance of the classifier.

2)RMSE: RMSE is an indicator used to measure the prediction accuracy of a prediction model on continuous data. It measures the root mean square difference between the predicted value and the true value, indicating the average deviation between the predicted value and the true value. It is one of the commonly used performance evaluation indicators in regression tasks.

$$RMSE = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(\widehat{y_i} - y_i)^2} \tag{1}$$

where $\widehat{y_i}$ is the actual value, y_i is the predicted value, and N is the number of samples.

3) MAE: MAE is the average of the absolute errors, which can better reflect the actual situation of the predicted value error. The calculation principle is to sum up the absolute value of the difference between the true value and the predicted value and then average it.

$$MAE = \frac{1}{N}\sum_{i=1}^{N}|\widehat{y_i} - y_i| \tag{2}$$

where $\widehat{y_i}$ is the actual value, y_i is the predicted value, and N is the number of samples.

B. Model framework

The DL model based on mixed molecular fingerprint, molecular fragment, and molecular graph is generally divided into a mixed fingerprint module (mFPN), a molecular graph module (GNN), a molecular fragment module (FGN) for feature extraction, and a comprehensive module (FCN) for feature-combined. As shown in Fig. 1, the model receives the SMILES

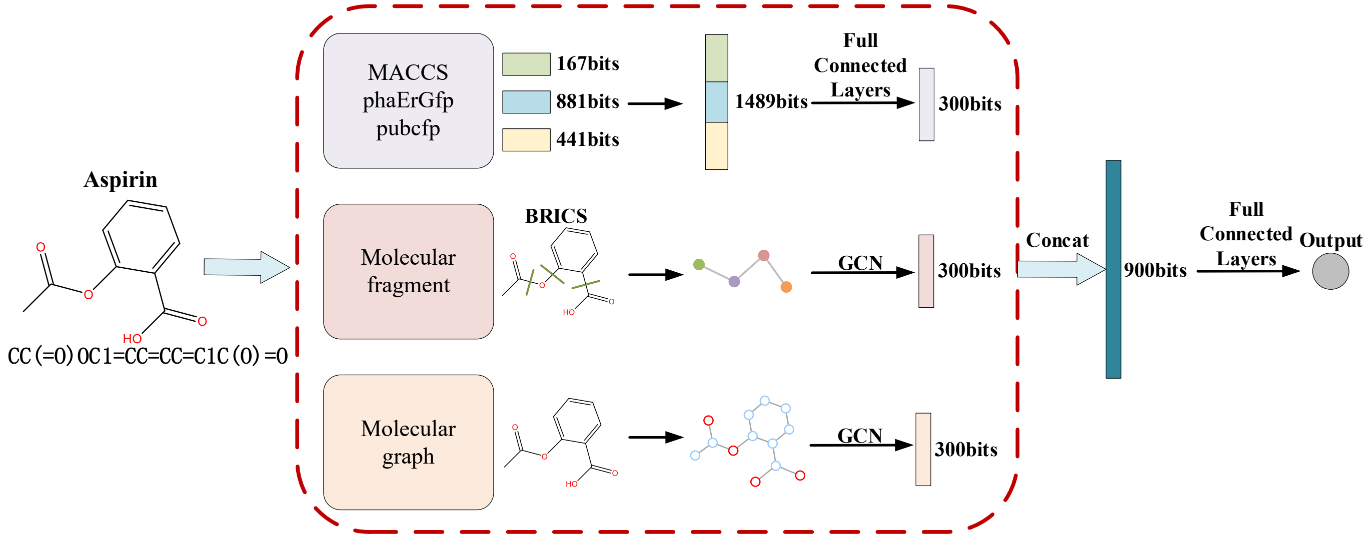

Fig. 1. Model architecture of PFG

representation of the molecule. Process it into mixed molecular fingerprint, molecular fragment structure data, and molecular graph structure data. Input to mFPN, FGN and GNN modules respectively. mFPN extracts a priori physicochemical and structural information predefined by chemists from molecular fingerprint. FGN extracts functional group information of molecules from molecular fragment. GNN automatically extracts the physical, chemical and structural features of molecules from molecular graph. FCN combines the features obtained in the FPN, FGN and GNN modules, and finally outputs the model's prediction results of molecular properties.

a)Module based on mixed molecular fingerprint: The mixed molecular fingerprint-based module takes molecular fingerprint as input. The substructure information, physical and chemical properties and pharmacophore information of molecules are extracted in the multi-layer fully connected neural network and output to the comprehensive module.

The mixed molecular fingerprint is obtained by combining the Molecular ACCess System fingerprint (MACCS) [15], Pharmacophore Extended Reduced Graph fingerprint (phaErGfp) [16] and PubChem fingerprint (pubcfp) [17]. The MACCS is a binary fingerprint used to represent molecular structure information. It is defined based on whether the molecule contains specific substructures. Each feature corresponds to a specific chemical substructure. If this feature exists in the molecule, the value of the binary bit corresponding to this feature is 1, otherwise it is 0. The MACCS used in this paper is 166+1 dimensions. The phaErGfp is a 2D pharmacophore fingerprint. It uses the Extended Reduced Graph method combined with pharmacophore type node descriptions to encode molecular features. The phaErGfp used in this paper is 881 dimensions. The pubcfp is a fingerprint based on molecular substructure keys, covering a large number of chemical structures. The pubcfp used in this paper is 441 dimensions. Three types of molecular fingerprints are spliced in the model, as shown in Equation (3). The spliced fingerprint is called a mixed fingerprint (mFP) and is 1489 dimensions.

$$mFP = MACCS + phaErGfp + pubcfp \tag{3}$$

The mFPN module uses a three-layer fully connected neural network to extract information from molecular fingerprint. The calculation of a single-layer fully connected neural network is shown in Equation (4). The mFP obtained by splicing is input into the fully connected layer. After the calculation of the hidden layer, the feature vector representing the molecular fingerprint is output.

$$Y = W * FP + b \tag{4}$$

where Y is the output of the fully connected layer, FP is the input of the fully connected layer, W is the network training weight in the fully connected layer, and b is the training bias in the fully connected layer.

b)Module based on molecular fragment: The molecular fragment-based module takes molecular fragment information as input and extracts the functional group information of the molecule in a two-layer graph convolutional neural network (GCN). After iterative updating, it is output to the comprehensive module.

Molecular fragment representation is a representation form that represents molecules into graphs based on fragments. Obtaining fragment-level molecular representation includes three parts, namely molecular fragmentation, generating fragment features, and extracting molecular information based on GCN. Given a molecular dataset, the molecules are first fragmented into fragments. Decompose molecules using the Breaking of Retrosyntheti-cally Interesting Chemical Substructures (BRICS) algorithm [18] that utilizes knowledge in the chemical domain. The BRICS algorithm develops 16 decomposition rules and breaks bonds in molecules that match a set of chemical reactions. After obtaining the molecular fragments, they are converted into graph structure data. Represent molecular fragments as nodes of the graph and the bonds connecting the fragments as edges of the graph. As shown in Tab. 2 , the fragment node feature is a 22-dimensional feature composed of the atom symbol in the fragment, the sum of H of the fragment, the valence of the fragment, the sum of formal charges of the fragment, aromaticity, the sum of atomic masses, and the sum of internal bonds. vector. The segment edge feature

TABLE II. FRAGMENT NODE FEATURE

Feature Type	***Feature size***	***Feature description***
Atomic features within fragments	16	Atomic symbol：B、C、N、O、F、Si、P、S、Cl、As、Se、Br、Te、I、At、other (*0.1)
Quantity of H	1	The H sum of the fragments (*0.1)
Valence	1	The valency of the fragment
Formal charge	1	The sum of the formal charges of a fragment
Aromaticity	1	Whether the fragment is aromatic
Total atomic mass	1	Sum of atomic masses (*0.01)
Number of bonds	1	Number of substructure covalent bonds: single bond 1, double bond 2, triple bond 3, directional bond 1.5 (*0.1)

is represented by the adjacency relationship matrix between segments, a 2-dimensional matrix composed of the starting node and the ending node.

Finally, the fragment node features and fragment edge features of the molecule are input into the GCN network model. The node information update formula of GCN is shown in Equation (5). As the number of iterations increases, the node receives more and more information. After the iteration is completed, the information of each node in the graph is summarized as a molecular fragment representation vector.

$$H^{(l+1)} = \sigma\left(\widetilde{D}^{-\frac{1}{2}}\tilde{A}\widetilde{D}^{-\frac{1}{2}}H^{(l)}W^{(l)}\right) \quad (5)$$

where $\tilde{A}$ is the adjacency matrix with added self-connection, $\widetilde{D}$ is the degree matrix of the graph node, $W^{(l)}$ is the weight matrix of the first layer of the neural network, $\sigma(\cdot)$ is the activation function, $H^{(l)}$ is the activation matrix of the first layer, and $H^{(0)}$ is a matrix composed of node feature vectors.

c)Module based on molecular graph: Molecular graph-based module takes as input the graph structure information of molecules. Molecular structural features and physicochemical information are extracted in a two-layer graph convolutional neural network (GCN). After iterative updating, it is output to the comprehensive module.

Molecular graph representation is the mapping of molecules into graph data $G(V, E)$ that can be interpreted by models. Among them, $V = \{a_1, a_2, \ldots, a_N\}$ represents the set of nodes in the graph, and the nodes are atoms. $E = \{e_1, e_2, \ldots, e_N\}$ represents a set of edges, which are chemical bonds. The node feature is a 22-dimensional vector consisting of the atomic symbol, the number of connected H's, valency, formal charge, aromaticity, atomic mass, and the number of connected bonds. The edge feature is represented by the adjacency relationship matrix between atoms in the molecule, a 2-dimensional matrix composed of starting nodes and ending nodes. Finally, the node features and edge features of the molecules are input into the GCN network model. Output a 300-dimensional feature vector, which is the expression form of molecular graph representation.

d)Comprehensive module: As shown in Fig.1, FPN, FGN, and GNN respectively output feature vectors representing molecular fingerprint, molecular fragment, and molecular graph. After receiving them, the comprehensive module FCN splices the three and inputs it into the three-layer fully connected feedforward neural network. The output is the final molecular property prediction result.

III. RESULTS AND DISCUSSION

The evaluation of the PFG model is approached from two perspectives. First, by comparing the single representation model, the impact of fingerprint-based module, fragment-based module and graph-based module in the model is analyzed. Evaluate whether molecular representation that combine fingerprint, molecular fragment, and graph can improve model performance. Then, the model is compared with current mainstream machine learning and DL algorithms on 10 public benchmark datasets to evaluate the model's prediction accuracy of molecular properties.

The model training process uses common techniques in DL training. The study uses the Adam optimizer to adaptively update the learning rate to alleviate the oscillation problem existing in gradient descent. Eliminate the randomness of model training by setting random seeds. Use Bayesian optimization strategy to find hyperparameter combinations to improve the efficiency and quality of hyperparameter optimization. The PFG model selected four hyperparameters for optimization. The hyperparameters of the model are shown in Tab. 3. (The optimized hyperparameter values shown are based on the BBBP dataset as an example)

The experimental environment is Windows operating system, and the hardware platform is 13th Gen Intel(R) Core(TM) i9-13900K, NVIDIA GeForce RTX 4090 Ti graphics

TABLE III. THE HYPER-PARAMETERS

Parameters	***Description***	***Value***
batch_size	The input batch_size	50
max epoch	The maximum number of rounds trained	30
init_lr	Initial learning rate	0.0001
max_lr	Maximum learning rate	0.003
final_lr	Final learning rate	0.0001
activation	Activation function. Used to enhance the nonlinear capabilities of the model.	ReLU
warmup_epochs	The cycle in which the initial learning rate rises linearly to the maximum learning rate	2
hidden_size	The hidden layer dimension of the fully connected layer Optimization scope:200-600	300
fp_2_dim	The dim of the second layer in fpn Optimization scope :128-640	512
hide_features	The hidden layer dimension of GCN Optimization scope:128-512	256
dropout	Dropout ratio. Used to enhance the generalization ability of the model. Optimization scope:0-0.6	0.1

processing unit, 64G memory server computer. The development tools are Python3.7.16 and PyTorch1.11.0.

A. Validation of combined representation method

To extract functional group information of molecular fragment and structural features of molecular graph, obtain a more comprehensive graphical representation. Model FG was established by combining molecular fragment and molecular graph representations, and compared with model F based on molecular fragment representation and model G based on molecular graph representation. The results were shown in Tab. 4. Model FG achieved better prediction results on 9 datasets. The effectiveness of combining molecular fragment with molecular graph representations was verified.

To evaluate the effectiveness of the molecular representation method that combines mixed molecular fingerprint and graph representation. The PFG model was compared with the model P based on molecular fingerprint representation and the model FG based on graph representation. The results were shown in Tab. 5. The PFG model achieved better prediction results on 9 datasets. It verified the effectiveness of the molecular representation method that combines molecular fingerprint with graph representation.

TABLE IV. RESULTS OF COMBINING FRAGMENT WITH GRAPH REPRESENTATION

dataset	*metric name*	*F*	*G*	*FG*
BBBP	ROC-AUC	0.871	0.896	**0.922**
BACE	ROC-AUC	0.720	0.774	**0.821**
ClinTox	ROC-AUC	0.910	0.883	**0.940**
SIDER	ROC-AUC	0.594	**0.648**	0.633
Tox21	ROC-AUC	0.767	0.76	**0.809**
ESOL	RMSE	0.871	1.263	**0.773**
Freesolv	RMSE	1.912	2.038	**1.627**
Lipo	RMSE	1.012	1.046	**0.965**
QM7	MAE	116.3	132.9	**107.4**
QM8	MAE	0.0269	0.0217	**0.0192**

TABLE V. RESULTS OF COMBINING FINGERPRINT WITH GRAPH REPRESENTATION

dataset	*metric name*	*P*	*FG*	*PFG*
BBBP	ROC-AUC	0.945	0.922	**0.971**
BACE	ROC-AUC	0.929	0.821	**0.936**
ClinTox	ROC-AUC	0.894	**0.940**	0.924
SIDER	ROC-AUC	0.690	0.633	**0.701**
Tox21	ROC-AUC	0.855	0.809	**0.858**
ESOL	RMSE	0.581	0.773	**0.553**
Freesolv	RMSE	0.683	1.627	**0.649**
Lipo	RMSE	0.531	0.965	**0.525**
QM7	MAE	69.7	107.4	**66.8**
QM8	MAE	0.0112	0.0192	**0.0103**

B. Comparative experiment of the PFG model

In the comparative experiment, the PFG model was compared with a variety of advanced machine learning and DL methods based on molecular fingerprint and graph. The FraGAT method comes from a fragment-oriented multi-scale graph attention network proposed by Zhang et al. The FP-GNN method comes from the molecular property prediction model based on molecular fingerprints and graph neural networks proposed by the team of Professor Wang Ling of South China University of Technology. The evaluation results on 4 types of public benchmark datasets were shown in Tab. 6. The PFG model used the same data splitting method as the other 9 comparison objects.

The Physiology and Toxicity (ADMET) dataset documents the effects of compounds in living organisms. Among the 10 datasets, the classification datasets ClinTox, BBBP, SIDER, and Tox21 were used to evaluate the predictive ability of PFG on ADMET properties. As shown in Tab. 6, the PFG model has achieved better prediction results on the 3 datasets BBBP, SIDER, Tox21, and ClinTox. It shows that the PFG model can accurately predict the physiological and toxicological properties of compounds. The PFG model can be applied to eliminate unsuitable molecules in early screening of drug discovery, helping to reduce the cost of new drug development.

The physical and chemical properties of a drug can reflect its pharmacokinetic stage in the body. It is closely related to the binding efficiency of targeted drugs and plays a key role in the development of drug candidates. Therefore, accurate prediction of the physicochemical properties of molecules is essential for drug discovery. In 10 datasets, the regression datasets FreeSolv, ESOL and Lipo were used to evaluate the prediction ability of PFG on physical and chemical properties. As shown in Tab. 6, the PFG model has achieved better prediction results on the datasets FreeSolv, ESOL and Lipo. It shows that the PFG model is effective in predicting the physical and chemical properties of molecules.

Bioactivity datasets collect binding affinity data between compounds and different biological targets. The presence or absence of biological activity determines whether a compound has research value. In 10 datasets, the classification dataset BACE was used to evaluate the model's predictive ability for biological activity. As shown in Tab. 6, the PFG model achieved better prediction results on the dataset BACE. It shows that the PFG model can accurately predict the biological activity of molecules.

The quantum mechanical properties of molecules can help understand the electronic structure and interactions of molecules, which is beneficial to understanding the interactions between drugs and biomolecules. This paper uses datasets QM7 and QM8 to evaluate the model's ability to predict quantum mechanical properties. As shown in Tab. 6, the PFG model achieved better prediction results on the QM8 dataset and suboptimal prediction results on the QM7 dataset. It shows that the PFG model can accurately predict the quantum mechanical properties of molecules.

The PFG model achieved better prediction results on the prediction task of 9 benchmark datasets, and achieved suboptim-

TABLEVI. COMPARISON OF EXPERIMENTAL MODEL PREDICTION RESULTS

dataset	*metric name*	*SVM*	*GBDT*	*RF*	*XGBoost*	*GAT*	*Attention FP*	*DMPNN*	*FraGAT*	*FP-GNN*	*PFG*
BBBP	ROC-AUC	0.711	0.914	0.924	0.921	0.859	0.866	0.932	0.924	0.948	**0.971**
BACE	ROC-AUC	0.811	0.895	0.905	0.907	0.633	0.834	0.843	0.874	0.927	**0.936**
ClinTox	ROC-AUC	0.582	0.912	0.841	0.900	0.828	0.906	0.821	0.881	0.904	**0.924**
SIDER	ROC-AUC	0.552	0.610	0.626	0.624	0.583	0.663	0.602	0.580	0.697	**0.701**
Tox21	ROC-AUC	0.642	0.776	0.790	0.732	0.806	0.851	0.809	0.759	0.856	**0.858**
ESOL	RMSE	0.913	0.755	0.811	0.798	1.198	0.631	0.888	0.575	0.600	**0.553**
Freesolv	RMSE	0.928	0.964	0.927	1.093	2.153	1.242	2.283	0.953	0.799	**0.649**
Lipo	RMSE	0.673	0.567	0.577	0.546	0.948	0.636	0.616	0.543	0.532	**0.525**
QM7	MAE	103.0	84.3	104.8	98.2	106.0	103.5	96.3	**56.1**	70.5	66.8
QM8	MAE	0.1308	0.1069	0.1069	0.1067	0.0174	0.0320	0.0143	0.0169	0.0108	**0.0103**

al prediction results on the prediction task of 1 benchmark dataset. It has shown relatively excellent results in the comprehensive evaluation compared with other comparative methods. This demonstrates the complementary nature of new molecular representation methods that combine mixed fingerprint, molecular fragment, and molecular graph. It can improve the information acquisition amount of the DL model and improve the model's prediction performance of molecular properties. The PFG model performed well on 10 public benchmark datasets, demonstrating PFG's competitiveness among DL methods applied to drug discovery.

IV. CONCLUSION

This paper first combined molecular fragment with molecular graph to obtain a more comprehensive graph representation. And further combined the mixed molecular fingerprint with the graph representation form to propose a multi-modal molecular representation method. The method was beneficial to extracting the physical and chemical properties, functional group information and structural features of molecules. Then a molecular property prediction model PFG was established based on the method, and the model was evaluated on 10 benchmark datasets. The effectiveness of the combined representation method was demonstrated. The PFG model was also compared with state-of-the-art models. Experimental results shown that the PFG model can achieve better prediction performance in most cases. In the future research, it is a promising direction to combine the 3D geometric structure information of molecules into molecular representation. 3D molecular diagrams represent atomic structures as a set of atoms and their 3D coordinates, including more spatial information about the atoms. Incorporating 3D molecular graph in molecular representation can provide missing geometric information. But 3D molecules are dynamic structures, as atoms are constantly moving in 3D space. Therefore, the 3D coordinates of molecules present challenges that are difficult to calculate and sometimes impossible to determine.

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